

Neck-Hook Fingertip: CAD Guide

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Why change the fingers

The current 2F-85 fingers hold a lifted bottle by friction alone, and a friction pinch on a round neck is a compliant pivot: the bottle can swing and tilt for many seconds with almost no damping instead of settling. The tilt, not just sideways drift, is what causes misses downstream (a second arm placing a bell over the cap on top of the bottle).

Measured on this project's beer bottle (473ml longneck, held 80mm above its own centre of mass):

- Peak tilt after a 50mm lift ranged from about 2 degrees to over 10 degrees across otherwise-identical grasps.
- One traced run never settled below 15 seconds of continuous oscillation.
- Squeezing harder made it worse, not better: 1.68 degrees of peak tilt at a light squeeze command, 5.92 degrees at the current default, 10.83 degrees at a heavier one. More pinch force means more spring in the joint the bottle hangs from, not less.

The fix already has a CAD generator

This project already has a parametric fingertip generator built for exactly this idea, at `fingertip/fingertip.py`, using [build123d](#) (Python CAD). It exports STEP for CAD review and STL for slicing.

Instead of two flat pads pinching a smooth section, each fingertip has a pocket that wraps part of the neck and steps outward above a ledge. The bottle's own collar ring rests on that ledge, so it hangs off a rigid mechanical stop instead of a squeezed friction contact -- the same principle as a factory conveyor carrying bottles by the neck. A rigid interlock has far less give than a compressed rubber-on-glass contact, which is the spring this pendulum problem needs removed, not damped.

It is unused in this project so far: it was built for general pick-and-place by the neck, and the beer-bottle-holding task that found the swing problem still uses the stock friction-pinch fingers.

The bottle's real geometry

Measured off this project's own 473ml/16oz longneck model

(`make_beer_and_opener.py`), by taking the widest point in each ring of mesh vertices, base at $z = 0$:

z range (mm)	feature	diameter
0-130	body	64.4mm
130-152	shoulder, full width	64.4mm
152-170	shoulder tapering -- the stop under today's grip	64.4 to 51.0mm
170-200	neck, straight -- where today's pads close	38.7mm
200-242	neck, tapering	38.7 to 23.3mm
242-258.1	crown finish: transfer bead at $z=244$, sealing lip at $z=253-257$	bead 27.3mm

The transfer bead is the collar to hook under: 27.3mm sitting above a 23.3mm neck just below it, a 2mm overhang on each side. That is the standard feature crown caps crimp onto, and it is why `fingertip.py` already lists "crown" as a named finish (`FINISHES` table) -- but with no dimensions filled in yet. These numbers are the ones to put there, pending the caliper check in the next section.

What is still missing

Two things the generator cannot supply on its own, both called out in its own README as blockers before printing anything that bolts on:

- 1. Robotiq 2F-85 mounting dimensions.** `mount_hole_spacing`, `mount_hole_d`, `mount_face_w`, `mount_face_h`, `mount_plate_t` in the `Params` dataclass are invented placeholders today -- not Robotiq's, not measured. Get them off the real fingertip (calipers) or Robotiq's official 2F-85 STEP file, and replace all five. The tool prints a loud warning naming whichever ones are still untouched.
- 2. A caliper check on a real bottle.** The 23.3mm / 27.3mm numbers above came off a simulation mesh, not a certified drawing or a real bottle. Confirm them (and the bead's axial position and profile) on an actual bottle from this bar's supplier before committing to a print run.

Step by step

1. `cd fingertip && python3 -m venv .venv && .venv/bin/python -m pip install -r requirements.txt`
2. Edit the `Params` dataclass in `fingertip.py`: fill in the five real mounting dimensions from the section above.
3. Fill in `FINISHES["crown"]` with `neck_d = 23.3`, `collar_d = 27.3` (mm, pending the caliper check), plus a `clearance` sized to your printer's tolerance.
4. `python fingertip.py --mount-check --out ./build` first -- a few minutes of filament, just the mounting plate, to confirm the bolt pattern before spending an hour on a full tip.
5. `python fingertip.py --neck-d 23.3 --collar-d 27.3 --out ./build` for the full pair once the plate fits.
6. `pytest` in `fingertip/` runs the geometry validation suite (a `pytest.ini` already disables the ROS 2 plugins this machine puts on `PYTHONPATH`).
7. `python preview.py` renders the exported STL to a PNG if you want a quick look without opening CAD.

Validation runs before every export and raises rather than silently adjusting a dimension -- a message naming the exact numbers that failed means the part it refuses to build would not have fit.

Printing

- **Material: TPU 95A.** Grippy enough that the neck does not skid in the pocket, tough enough in layer adhesion to carry a full bottle.
- **Orientation: mounting face flat on the bed.** The ledge is a horizontal load-bearing surface; printed standing up, the load peels layers apart in tension. Mounting-face-down turns the layer planes 90 degrees so the load is carried in shear instead. Expect to need supports in the pocket at this orientation -- that surface is not load-bearing, so a scarred finish there costs nothing.
- **Walls: 4-5 perimeters** (about 2mm at a 0.4mm nozzle). In TPU the perimeters do essentially all the work; 40-60% gyroid infill behind them is plenty.
- **Other settings:** 0.2mm layers, 20-30mm/s, no or minimal retraction, dry filament -- TPU absorbs moisture fast and prints stringy and weak when wet.

Open risks

- **Bearing surface is shallow.** A 2mm-per-side overhang is modest -- confirm it carries 700g (this bottle's measured mass, liquid included) without slipping off the ledge, especially under any vibration from the arm's own motion.
- **The press load path changes.** Today's friction pinch absorbs some of the lateral push a second arm applies while pressing a cap; a rigid hook carries that load differently. Check clearance against the existing press travel and cap-release step before assuming the swap is drop-in.
- **Confirm the fix actually works before switching over.** Print, mount, and re-run this project's own tilt trace (position + orientation logging on the held bottle) to check peak tilt actually drops -- a rigid hook can still allow some play at the neck-to-ledge contact.
- **Numbers are sim-derived, not certified.** Re-measure the real bottle before finalizing dimensions that go into a printed or machined part.